

FinRL (Financial Reinforcement Learning)

FinRL is an open-source framework designed for financial reinforcement learning. It's built to help users develop stock trading strategies using deep reinforcement learning (DRL) algorithms. The framework is structured in three layers: market environments, agents, and applications.

OBJECTIVE:

- The primary goal of FinRL is to develop a trading strategy that can adapt and optimize performance in real-time market conditions.
- FinRL can be used to build automated trading bots that make decisions based on learned strategies.
- Reinforcement learning can help in optimizing asset allocation and managing investment portfolios.
- FinRL can assist in identifying and mitigating financial risks through adaptive strategies.

Dataset Collection for Stock Trading Bot

- Identify Data Sources: Use reliable sources like Yahoo Finance, Alpha Vantage, or Quandl
- Preprocess Data: Clean and preprocess the data to handle missing values, outliers, and normalize the data if necessary

Build a Stock Trading Bot :

- Define Strategy: Decide on the trading strategy to implement (e.g., momentum trading, mean reversion).
- Define the Trading Environment: Configure the StockTradingEnv with initial capital, transaction costs, and reward functions.
- Train the Agent: Select a reinforcement learning algorithm . Train the agent on the environment.
- Backtest the Bot: Test the bot on historical data to evaluate its performance
- Deploy the Bot: Once satisfied with the performance, deploy the bot to trade with real money

```
# Install Python 3.10 and set it as the default
```

```
!sudo apt-get update -y
```

```
!sudo apt-get install python3.10 python3.10-distutils python3.10-dev -y
```

```
# Update alternatives to use Python 3.10
```

```
!sudo update-alternatives --install /usr/bin/python3 python3 /usr/bin/python3.10 1
```

```
!sudo update-alternatives --config python3
```

```
# Verify the Python version
```

```
!python3 --version
```

```
# Install pip for Python 3.10
```

```
!sudo apt-get install python3.10-venv -y
```

```
!curl -sS https://bootstrap.pypa.io/get-pip.py | python3.10
```





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Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
Downloading pip-25.0-py3-none-any.whl (1.8 MB)
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Downloading setuptools-75.8.0-py3-none-any.whl (1.2 MB)
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Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Installing collected packages: wheel, setuptools, pip
```

```
!pip install finrl
```



```
Collecting finrl
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  Downloading FinRL-0.3.7-py3-none-any.whl.metadata (909 bytes)
  Downloading FinRL-0.3.7-py3-none-any.whl (127 kB)
Installing collected packages: finrl
Successfully installed finrl-0.3.7
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```
!pip install finrl stable-baselines3[extra] yfinance stockstats matplotlib
```



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Requirement already satisfied: finrl in /usr/local/lib/python3.10/dist-packages (6.0.0)
Collecting yfinance
  Downloading yfinance-0.2.52-py2.py3-none-any.whl.metadata (5.8 kB)
Collecting stockstats
  Downloading stockstats-0.6.3-py2.py3-none-any.whl.metadata (39 kB)
Collecting matplotlib
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Collecting stable-baselines3[extra]
  Downloading stable_baselines3-2.4.1-py3-none-any.whl.metadata (4.5 kB)
Collecting gymnasium<1.1.0,>=0.29.1 (from stable-baselines3[extra])
  Downloading gymnasium-1.0.0-py3-none-any.whl.metadata (9.5 kB)
Collecting numpy<2.0,>=1.20 (from stable-baselines3[extra])
  Downloading numpy-1.26.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (2.5 kB)
Collecting torch>=1.13 (from stable-baselines3[extra])
  Downloading torch-2.5.1-cp310-cp310-manylinux1_x86_64.whl.metadata (28 kB)
Collecting cloudpickle (from stable-baselines3[extra])
  Downloading cloudpickle-3.1.1-py3-none-any.whl.metadata (7.1 kB)
Collecting pandas (from stable-baselines3[extra])
  Downloading pandas-2.2.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18.3 kB)
Collecting opencv-python (from stable-baselines3[extra])
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Collecting pygame (from stable-baselines3[extra])
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Collecting tensorboard>=2.9.1 (from stable-baselines3[extra])
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Collecting psutil (from stable-baselines3[extra])
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Collecting tqdm (from stable-baselines3[extra])
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Collecting rich (from stable-baselines3[extra])
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Collecting ale-py>=0.9.0 (from stable-baselines3[extra])
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Collecting pillow (from stable-baselines3[extra])
  Downloading pillow-11.1.0-cp310-cp310-manylinux_2_28_x86_64.whl.metadata (9.1 kB)
Collecting requests>=2.31 (from yfinance)
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Collecting multitasking>=0.0.7 (from yfinance)
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Collecting lxml>=4.9.1 (from yfinance)
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  Downloading pytz-2024.2-py2.py3-none-any.whl.metadata (22 kB)
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Collecting peewee>=3.16.2 (from yfinance)
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----- 948.2/948.2 kB 23.6 MB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Collecting beautifulsoup4>=4.11.1 (from yfinance)
  Downloading beautifulsoup4-4.12.3-py3-none-any.whl.metadata (3.8 kB)
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Collecting fonttools>=4.22.0 (from matplotlib)
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Collecting kiwisolver>=1.3.1 (from matplotlib)
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Collecting packaging>=20.0 (from matplotlib)
  Downloading packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/lib/python3/dist-packages (from matplotlib)
Collecting python-dateutil>=2.7 (from matplotlib)
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Collecting typing-extensions (from ale-py>=0.9.0->stable-baselines3[extra])
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Collecting soupsieve>1.2 (from beautifulsoup4>=4.11.1->yfinance)
  Downloading soupsieve-2.6-py3-none-any.whl.metadata (4.6 kB)
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Requirement already satisfied: six>=1.9 in /usr/lib/python3/dist-packages (from farama-notifications)
Collecting webencodings (from html5lib>=1.1->yfinance)
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Collecting tzdata>=2022.7 (from pandas->stable-baselines3[extra])
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  Downloading Markdown-3.7-py3-none-any.whl.metadata (7.0 kB)
Collecting protobuf!=4.24.0,>=3.19.6 (from tensorboard>=2.9.1->stable-baselines3[extra])
  Downloading protobuf-5.29.3-cp38-abi3-manylinux2014_x86_64.whl.metadata (592 bytes)
Requirement already satisfied: setuptools>=41.0.0 in /usr/local/lib/python3.10/dist-packages
Collecting tensorboard-data-server<0.8.0,>=0.7.0 (from tensorboard>=2.9.1->stable-baselines3[extra])
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Collecting filelock (from torch>=1.13->stable-baselines3[extra])
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Collecting networkx (from torch>=1.13->stable-baselines3[extra])
  Downloading networkx-3.4.2-py3-none-any.whl.metadata (6.3 kB)
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Collecting nvidia-cufft-cu12==11.2.1.3 (from torch>=1.13->stable-baselines3[extra])
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Collecting triton==3.1.0 (from torch>=1.13->stable-baselines3[extra])
Downloading triton-3.1.0-cp310-cp310-manylinux_2_17_x86_64_manylinux2014_x86_64
Collecting sympy==1.13.1 (from torch>=1.13->stable-baselines3[extra])
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Collecting mpmath<1.4, >=1.1.0 (from sympy==1.13.1->torch>=1.13->stable-baselines3[extra])
```

Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)
Collecting markdown-it-py>=2.2.0 (from rich->stable-baselines3[extra])
Downloading markdown_it_py-3.0.0-py3-none-any.whl.metadata (6.9 kB)
Collecting pygments<3.0.0,>=2.13.0 (from rich->stable-baselines3[extra])
Downloading pygments-2.19.1-py3-none-any.whl.metadata (2.5 kB)
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Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Collecting MarkupSafe>=2.1.1 (from werkzeug>=1.0.1->tensorboard>=2.9.1->stable-baselines3[extra])
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Downloading yfinance-0.2.52-py2.py3-none-any.whl (108 kB)
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Downloading matplotlib-3.10.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (10.0 MB)
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2.1/2.1 MB 62.0 MB/s eta 0:00:00
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Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.55.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.6 MB)
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Downloading gymnasium-1.0.0-py3-none-any.whl (958 kB)
958.1/958.1 kB 34.5 MB/s eta 0:00:00
Downloading cloudpickle-3.1.1-py3-none-any.whl (20 kB)
Downloading html5lib-1.1-py2.py3-none-any.whl (112 kB)
Downloading kiwisolver-1.4.8-cp310-cp310-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (1.1 MB)
1.6/1.6 MB 52.1 MB/s eta 0:00:00
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Downloading pandas-2.2.3-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
13.1/13.1 MB 102.9 MB/s eta 0:00:00
Downloading pillow-11.1.0-cp310-cp310-manylinux_2_28_x86_64.whl (4.5 MB)
4.5/4.5 MB 80.1 MB/s eta 0:00:00
Downloading platformdirs-4.3.6-py3-none-any.whl (18 kB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2024.2-py2.py3-none-any.whl (508 kB)
Downloading requests-2.32.3-py3-none-any.whl (64 kB)
Downloading tensorboard-2.18.0-py3-none-any.whl (5.5 MB)
5.5/5.5 MB 89.3 MB/s eta 0:00:00
Downloading torch-2.5.1-cp310-cp310-manylinux1_x86_64.whl (906.4 MB)
906.4/906.4 MB 26.6 MB/s eta 0:00:00
Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl (363.4 MB)
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Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
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Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (24.6 MB)
24.6/24.6 MB 33.0 MB/s eta 0:00:00
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Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl (127.9
127.9/127.9 MB 49.1 MB/s eta 0:00:00
Downloading nvidia_cusparses_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl (207
207.5/207.5 MB 66.8 MB/s eta 0:00:00
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Downloading nvidia_nvtx_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (99 kB)
Downloading sympy-1.13.1-py3-none-any.whl (6.2 MB)
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Downloading triton-3.1.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
209.5/209.5 MB 64.7 MB/s eta 0:00:00
Downloading opencv_python-4.11.0.86-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_
63.0/63.0 MB 54.9 MB/s eta 0:00:00
Downloading psutil-6.1.1-cp36-abi3-manylinux_2_12_x86_64.manylinux2010_x86_64.many
Downloading pygame-2.6.1-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
14.0/14.0 MB 93.8 MB/s eta 0:00:00
Downloading rich-13.9.4-py3-none-any.whl (242 kB)
Downloading stable_baselines3-2.4.1-py3-none-any.whl (183 kB)
Downloading tqdm-4.67.1-py3-none-any.whl (78 kB)
Downloading absl_py-2.1.0-py3-none-any.whl (133 kB)
Downloading certifi-2024.12.14-py3-none-any.whl (164 kB)
Downloading charset_normalizer-3.4.1-cp310-cp310-manylinux_2_17_x86_64.manylinux20
Downloading Farama Notifications-0.0.4-nv3-none-any.whl (2.5 kB)
```

```
Downloading grpcio-1.70.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (5.9 MB)  
5.9/5.9 MB 103.5 MB/s eta 0:00:00  
Downloading idna-3.10-py3-none-any.whl (70 kB)  
Downloading Markdown-3.7-py3-none-any.whl (106 kB)  
Downloading markdown_it_py-3.0.0-py3-none-any.whl (87 kB)  
Downloading protobuf-5.29.3-cp38-abi3-manylinux2014_x86_64.whl (319 kB)  
Downloading pygments-2.19.1-py3-none-any.whl (1.2 MB)  
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Downloading soupsieve-2.6-py3-none-any.whl (36 kB)  
Downloading tensorboard_data_server-0.7.2-py3-none-manylinux_2_31_x86_64.whl (6.6 MB)  
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Downloading typing_extensions-4.12.2-py3-none-any.whl (37 kB)  
Downloading tzdata-2025.1-py2.py3-none-any.whl (346 kB)  
Downloading urllib3-2.3.0-py3-none-any.whl (128 kB)  
Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)  
Downloading filelock-3.17.0-py3-none-any.whl (16 kB)  
Downloading fsspec-2024.12.0-py3-none-any.whl (183 kB)  
Downloading jinja2-3.1.5-py3-none-any.whl (134 kB)  
Downloading networkx-3.4.2-py3-none-any.whl (1.7 MB)  
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Downloading webencodings-0.5.1-py2.py3-none-any.whl (11 kB)  
Downloading MarkupSafe-3.0.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (17 kB)  
Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)  
Downloading mpmath-1.3.0-py3-none-any.whl (536 kB)  
536.2/536.2 kB 20.6 MB/s eta 0:00:00  
Building wheels for collected packages: peewee  
Building wheel for peewee (pyproject.toml) ... done  
Created wheel for peewee: filename=peewee-3.17.8-cp310-cp310-linux_x86_64.whl
```

RESTART SESSION

```
import sys
print(sys.executable)
```

 /usr/bin/python3

```
!pip install --upgrade --no-cache-dir finrl
```

 Requirement already satisfied: finrl in /usr/local/lib/python3.10/dist-packages (0.3.



```
import sys
sys.path.append("/path/to/finrl")
```

```
# Install FinRL and dependencies
```

```
!pip install finrl stable-baselines3[extra] yfinance matplotlib
```



```
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.10/dist-packages (peewee==3.16.2)
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```

Encountered the error while importing the libraries even though the FinRL package is already installed.

```
# Install FinRL and dependencies using !pip in the correct environment
!pip install --upgrade --no-cache-dir finrl stable-baselines3[extra] yfinance matplotlib

# Import required libraries
from finrl.preprocessing.data import DataProcessor

# Initialize DataProcessor with Yahoo Finance as the data source
dp = DataProcessor(data_source="yahoofinance")

# Download stock data for AAPL and MSFT from 2020 to 2021
data = dp.download_data(
    ticker_list=["AAPL", "MSFT"],
    start_date="2020-01-01",
    end_date="2021-01-01",
    time_interval="1D"
)

# Display the first few rows of the dataset
```

```
print(data.head())
```



Requirement already satisfied: finrl in /usr/local/lib/python3.10/dist-packages (0.3.
Requirement already satisfied: yfinance in /usr/local/lib/python3.10/dist-packages (0.
Requirement already satisfied: matplotlib in /usr/local/lib/python3.10/dist-packages
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Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/local/lib/python3.10/dist-packages/MarkupSafe-2.1.1
```

ModuleNotFoundError Traceback (most recent call last)

[<ipython-input-19-bc19c23b6915>](#) in <cell line: 0>()
3

3

4 # Import required libraries

----> 5 from finrl.preprocessing.data import DataProcessor

6

7 # Initialize DataProcessor with Yahoo Finance as the data source

ModuleNotFoundError: No module named 'finrl'